INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

COLD FOOD COUNTER, MECHANICALLY REFRIGERATED (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers self-service, mechanically refrigerated, cold food counters with refrigerated storage compartment. The cold food counters are intended to be used in Naval shipboard food serving lines.

2. CLASSIFICATION

Type 1: Single-sided access
Type 2: Double-sided access

Style 1: With display case accessory (nonrefrigerated)
Style 2: With overhead breath guard accessory

Size 1: Three food storage pan capacity
Size 2: Four food storage pan capacity
Size 3: Five food storage pan capacity
Size 4: Six food storage pan capacity

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The cold food counter shall be a cabinet-type self-contained unit consisting of a recessed refrigerated cold pan counter top, air-cooled electrical refrigeration system, and refrigerated storage compartment with one or more doors. The cold food counter shall include a breaker strip, tray rails, and stainless steel food pans and adapter bars. Depending on the style selected, it shall also include a display case or an overhead breath guard. Size 1 counters shall be provided with a single door, while sizes 2, 3, and 4 shall be provided with two or more doors. Each storage compartment shall have two fixed shelves. Each cold food counter cold pan shall also be provided with two equal-size stainless steel covers to close the recessed opening for sanitation and chilling purposes.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson-Davis Highway, Arlington, VA 22242-5160.

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3.2 Cabinet body. The cabinet body shall have metal frame construction. The structural members shall have a thickness of 16-gauge stainless steel. Unless otherwise specified, the thickness of the exterior sides and front, excluding the cabinet top, shall be 18-gauge stainless steel. The thickness of the cabinet top shall be 16-gauge stainless steel. The counter top recessed cold pan opening shall accommodate three, four, five, or six full-size deep pans. The recessed opening shall be considered a food zone. The food pan flanges shall overlap the surface of the opening. Drains shall be sufficiently recessed such that food pans do not contact drain stoppers. A drain hose shall be provided and attached to the drain pipe and shall be of sufficient length to prevent draining over the condensing unit. Combinations of pans and adapter bars shall completely cover the recessed countertop opening.

3.3 Doors. Cold food counter doors shall be self-closing with built-in dwell when opened up to 100° (from the closed position), and shall have positive safety latching stops at 120°. Units shall be equipped with a positive door latch mechanism for the fully open and closed positions. Hinges and door pulls shall be chrome-plated or stainless steel. Each door shall have a one-piece replaceable gasket. Each door opening shall have an odorless, nontoxic, safety beveled, vinyl breaker strip with no exposed screws or fasteners. Each door corner shall be provided with a stainless steel fillet. The perimeters of openings shall be provided with concealed dew point compensators.

3.4 Mounting. The cold food counter shall be provided with four 6-inch high stainless steel legs suitable for secure attachment to the deck. (Shipboard requirement, see 7.2)

3.5 Storage compartment. Each storage compartment shall be provided with two fixed shelves, provisions shall be provided to allow shelves to be removable for servicing the evaporator section. A flush-mounted thermometer incorporating a range of 30 to 50°F shall be provided and shall be externally mounted at the front of the storage compartment on the door mullion, but shall not be mounted on the door.

3.6 Tray rails. Unless otherwise specified, a drop-type tray rail consisting of three 1-inch diameter tubes shall be supplied on the patron's side. When specified, a fixed tray rail shall be provided. The tray rail shall be located for single-sided or double-sided service, as required for the type of food counter. The tray rail shall extend the full length of the cabinet. The ends of each rail tube shall be closed. The height of the tray rail and brackets shall be adjustable between 34 and 35 inches, and shall be stainless steel. The tray rail and brackets shall also be capable of supporting the following weights, equally distributed over its full length.

Size 1: 75 pounds
Size 2: 100 pounds
Size 3: 125 pounds
Size 4: 150 pounds

3.7 Display case and protector shelf.

3.7.1 Style 1, cold food counter display case assembly. The style 1 cold food counter display case accessory assembly shall be located on the patron's side and shall consist of: a square or round stainless steel tube framework; two full length stainless steel or Plexiglas 1/4-inch shelves; and two Plexiglas breath guards, adjustable from closed vertical to 60° open. When specified, breath guards shall be provided on both sides of the display case. Breath guards shall be removable without tools. The legs of the display stand shall be attached to the countertop by at least two studs, bolts, or screws with lockwashers and nuts drawn up tight. When sleeves or collars are used,

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legs shall be inserted through the table top into sleeves or collars secured to the framework under the countertop. When bolts or screws are used, the fastener shall be attached to the framework. Both ends of the display case shall be provided with a 1/4-inch Plexiglas enclosure.

3.7.2 Style 2, overhead breath guard. The style 2, overhead breath guard accessory units are specified for self-service operations. The overhead breath guard shall consist of a stainless steel tubular framework with a one-piece Plexiglas 1/4-inch thick shelf and breath guard to protect both sides of the unit. The overhead breath guard shall be attached to the countertop by at least two studs, bolts, or screws with lock washers and nuts. Both ends of the overhead breath guard shall be open with the breath guard extending over the food product in the cold pan for protection. The breath guard shall be of sufficient height to allow placement and removal of glass racks having dimensions of 20 inches wide by 20 inches deep by 7 inches high. Cold food counters with styled superstructures shall be equipped with tray rails on both sides of the cold food counters for double service.

3.8 Food pans. Cold food counters shall be provided with rectangular unperforated pans and adapter bars of the sizes and quantities listed in Table 1. The countertop opening shall be completely covered when any combination of full size or 1/3-size pans and adapter bars are inserted in the normal use position for either three-pan, four-pan, five-pan, or six-pan capacity cold food counters.

TABLE 1. Pans and adapter bars.

Pan and adapter bar sizes	Pan size (2)	Size 1 quantity	Size 2 quantity	Size 3 quantity	Size 4 quantity
Full size, regular (4 inch depth)	Size 3	1	2	2	3
Full size, deep (6 inch depth)	Size 4	2	2	3	3
1/3-size	12	3	3	3	3
Adapter bars (1) (minimum length 21 inches)	---	2	3	4	5

Notes:
1. Each end flanged not less than 1/2 inch to rest on countertop
2. Pan size to be in accordance with A-A-50179

3.9 Refrigeration system. The refrigeration system shall consist of a compressor, compressor motor, air-cooled condenser, expansion device, evaporator, high and low pressure control or a high pressure and thermostatic control for automatic operation, and accessories. The thermostat shall be adjustable and shall be set and adjusted for a cut-out point of 38°F and a cut-in point of 44°F. The motor and compressor shall be hermetically sealed by either thermal or mechanical means. Electric motors shall be in accordance with NEMA MG-1. Each side of the counter condensing unit shall be provided with a removable louvered panel.

3.9.1 Evaporator. The evaporator shall be provided with one flat smooth surface forming the bottom of the cold food counter cold pan. A 1-inch diameter drain line shall be provided through the evaporator surface or cold plate. A rubber or plastic drain stopper shall be provided.

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3.9.2 Refrigerant (ozone depleting substance). The refrigerant shall be in accordance with ASHRAE 15, Group A1 (nonflammable), be in accordance with ARI 700 (purity), and be approved by the Environmental Protection Agency under the Significant New Alternate Program. The refrigerants permitted for use, in order of preference, shall be of one of the following: R134A, R-401A, or R-401B.

3.9.3 Performance. The cold food pan refrigeration system shall reduce the temperature of water in the food storage pans by 6°F within one hour with the initial water within a range of 50°F to 40°F. The cold food counter shall be filled to capacity with water. The refrigeration system shall maintain the temperature of water in the food storage pans at a temperature not greater than 44°F in an ambient air temperature surrounding the unit of not less than 100°F for 4 hours. The running time shall not exceed 75 percent and the condensing unit motor shall not cycle on the motor protector. The exterior surfaces of the cold food counters shall be not greater than 10°F below the ambient air temperature surrounding the unit. The operating system shall be such that food products or liquids in the pans do not freeze or show any evidence of ice formation after a period of 4 hours of continuous normal cycle operation while the cold food counter's opening is completely covered with the combination of pans and adapter bars.

3.9.4 Condenser ventilation. Ventilation of the condensing unit shall be made by removable louvered panels. The ventilating panels on both the operator's side and patron's side shall be constructed of the same stainless steel material as the adjacent body components. The bottom of the condensing unit compartment shall also be ventilated.

3.10 Electrical requirements. The unit shall operate on 115 VAC, 60 Hz, single-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. Mercury-type float switches shall not be used. (Shipboard requirement, see 7.2)

3.11 Insulation installation. The countertop shall be completely filled with foamed-in-place insulation in accordance with ASTM C 1029. Ozone-depleting substances to propel the insulation shall not be used during manufacturing. The doors and walls of the refrigerated cabinet shall be insulated with not less than 2 inches of rigid insulation or foamed-in-place insulation. Rigid insulation, when used, shall have the layers bonded to each other with a minimum of joints, carefully filled and staggered. Insulation shall be bonded to the inner surfaces of the inner liner and outer shell by a commercial grade permanent cement. When copper tube is used for the evaporator, voids between the tubes shall be filled with a thermosetting mastic.

3.12 Inclined operation. The unit shall operate satisfactorily, in accordance with the requirements of this CID (such as the door remaining closed to prevent loss of contents), when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

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3.13 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.14 Physical size limitations. The unit shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8 inch radius corners. When specified for submarines, the unit shall pass or shall disassemble so that it can pass through a 30 inch diameter submarine access. The refrigeration system shall not require disassembly to pass through access openings. The width of the cold food counter shall be not greater than 54, 66, 78, and 90 inches for the three-pan, four-pan, five-pan, and six-pan sizes, respectively. The depth without tray rails and the height without legs shall be not greater than 25-1/2 inches. (Shipboard requirement, see 7.2)

3.15 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.16 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 2, NSF 7, UL 62, UL 207, and UL 471.

3.17 Finish. Stainless steel shall have a type 3 or smoother finish, in accordance with ASTM A 480.

3.18 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.18.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, and serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.18.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such ompliance.

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6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Type, style, and size required
- When required, fixed tray rails
- When required, tray rails located on the patron side or both the patron and attendant side
- When required, breath guards for both sides of the display case
- When required, manuals shall be in accordance with ASTM F 760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)
MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating

7.3.2 American National Standards Institute (ANSI) Standards. ANSI Standards are available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

ANSI C73A - Dimensions of Attachment Plugs and Receptacles for Marine Ships

7.3.3 Air-Conditioning & Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700 - Specifications for Fluorocarbons and Other Refrigerants

7.3.4 American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standards. ASHRAE Standards are available from the American Society of Heat, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle, N.E., Atlanta, GA 30329-2305.

AHSRAE 15 - Safety Code for Mechanical Refrigeration

7.3.5 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

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ASTM A 269 - Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels
ASTM A 366 - Steel, Sheet, Carbon, Cold-Rolled, Commercial Quality
ASTM A 480 - General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM C 1029 - Spray-Applied Rigid Cellular Polyurethane Thermal Insulation
ASTM F 760 - Food Service Equipment Manuals

7.3.6 National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from the National Electrical Manufacturers Association, 2101 L Street Northwest, Washington, DC 20037.

NEMA MG-1 - Motors and Generators
NEMA WD-1 - General Requirements for Wiring Devices

7.3.7 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 2 - Food Service Equipment
NSF 7 - Food Service Refrigerators and Storage Freezers

7.3.8 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 62 - Flexible Cord and Fixture Wire
UL 207 - Refrigerant-Containing Components and Accessories, Nonelectrical
UL 471 - Commercial Refrigerators and Freezers

7.3.9 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

Bayonne Stainless Products
98 Avenue E
Bayonne, NJ 07002-2526

Continental Refrigeration
Div/National Refrigeration and Air Conditioning Products
411 Yerkes Road
King of Prussia, PA 19406

Cospolich Refrigerator Co.
949 Industry Rd.
Kenner, LA 70062

Foster Refrigeration Corporation
N 2
Hudson, NY 12534

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Hobart Corporation Executive Offices
701 Ridge Avenue
Troy, OH 45374-3440

Keyser Stainless Corporation
Industrial Pk. at Waxler Road
Rt. 3 Box 445
Keyser, WV 26726
Low Temp Mfg. Company
P.O. Box 795
Jonesboro, GA 30236

RTF Manufacturing
RD 1 Route 66
Hudson, NY 12534

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Custodian:
Navy - SH
Army - GL

Preparing activity:
Navy - SH
(Project 7310-0889)

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